ChatGPT

Leakage-safe form intelligence layer for Australian horse racing markets

Betfair runner metadata and historical backfill feasibility

The Betfair Exchange “static market” surface is `listMarketCatalogue`, which returns **published** markets with status `ACTIVE` or `SUSPENDED` (not closed/settled markets). [1] This matters operationally because you cannot rely on querying `listMarketCatalogue` after a race to reconstruct what was known pre-race; you must persist point-in-time snapshots before the market turns in-play and closes. [1]

`RUNNER_METADATA` is a `MarketProjection` option that populates a **metadata map** for each runner when available. [2] Betfair’s documentation enumerates the horse-racing metadata keys that may appear, including:

- `runnerId`, `JOCKEY_NAME`, `TRAINER_NAME`, `OWNER_NAME` [1]
- demographics and breeding: `AGE`, `SEX_TYPE`, `COLOUR_TYPE`, `BRED`, `SIRE_NAME`, `DAM_NAME`, `DAMSIRE_NAME`, plus year-bred fields [1]
- race-day fields: `STALL_DRAW` (barrier), `CLOTH_NUMBER`, `WEIGHT_VALUE`, `WEIGHT_UNITS`, `JOCKEY_CLAIM`, `DAYS_SINCE_LAST_RUN`, and `WEARING` [1]
- limited “form” and ratings: `FORM` (a compact recent-form string), `OFFICIAL_RATING`, and `ADJUSTED_RATING` [1]

The same documentation is explicit that horse-racing metadata is **“when available”**, implying non-trivial missingness that can vary by jurisdiction/meeting and by individual runner. [1]

Coverage/completeness for AU markets

Public documentation does **not** publish AU-specific completeness percentages per key; in practice you should treat coverage as an empirical property you must measure with your own sampling and monitoring. [1] A Betfair-authored tutorial aimed at modelling confirms that `RUNNER_METADATA` can include trainer, jockey, age, and a class rating (among other fields), but also highlights the operational gap that the historical exchange stream files do **not** include `RUNNER_METADATA`. [3]

Where Betfair’s own tutorial demonstrates extracting Australian thoroughbred race metadata through a separate (explicitly described as “unsupported”) Hub endpoint, it notes that `jockeyName` is **not always populated**, and illustrates missing `weather` and `trackCondition` values in example outputs. [4] This is not a definitive measurement of `listMarketCatalogue` completeness for AU, but it is strong evidence that “race card” metadata fields can be missing in real production feeds even when they exist conceptually. [5]

Practical implication for the leakage-safe design: treat runner metadata as a partially observed feature block with explicit missingness handling, and build a daily automated completeness report by key (overall and stratified by state/track/meeting type) to keep the model stable when coverage drifts. [6]

Can Betfair runner metadata be backfilled from historic archives?

Betfair's exchange historical data service provides time-stamped streaming market data for purchase; the published specification indicates Australian/New Zealand market data availability from **October 2016** onward. [7] The historical feed describes `marketDefinition` and `RunnerDefinition` fields such as runner `id` (selectionId), `name`, `hc` (handicap), and `adjustmentFactor`, but does not describe the richer per-runner `RUNNER_METADATA` map (jockey/trainer/age/etc.) as part of the stream payload. [8]

Betfair's modelling tutorial states the same conclusion more directly: historical stream files do not include `RUNNER_METADATA`, and therefore you cannot extract it from those archives if you did not capture it prospectively. [3]

Betfair's own Hub education page describes three historical stream tiers (basic/advanced/pro) and their time resolution, but still frames the stream archive as odds/volume time-series rather than a source of rich runner form metadata. [9]

Net assessment:
- **From official Betfair historical stream archives alone:** runner metadata needed for "form intelligence" is **not backfillable** in a leakage-safe way, because it is absent from the archived stream and `listMarketCatalogue` does not serve closed markets for retrospective queries. [10]
- **Prospective capture:** feasible by persisting a T-10 (and optionally earlier) `listMarketCatalogue(RUNNER_METADATA)` snapshot per market and runner. [1]
- **Alternative Betfair endpoints:** the Betfair tutorial documents Hub endpoints (`/hub/racecard`, `/hub/raceevent`) as a workaround for historical race metadata, but it explicitly calls them "unsupported". [3] Any automated use also intersects with Betfair's general terms that prohibit screen scraping and "automated or manual collection of Betfair Data" without consent. [11] This combination makes Hub-based backfill a **licensing and stability risk** rather than a core design dependency. [12]

External data sources, coverage, and licensing constraints

A leakage-safe form layer needs at least: (i) stable horse/jockey/trainer identifiers and histories, (ii) track/distance/surface and (iii) performance indicators (sectionals, times, pace, etc.). Market-only Betfair features do not supply those histories, and the Betfair `FORM` string is too compressed for robust modelling. [1]

Data availability matrix

The matrix below focuses on sources that are (a) technically capable of supporting modelling and (b) explicit enough on licensing to evaluate risk.

Feature block	Source endpoint/provider	Historical depth	Missingness & latency	Licensing risk for model training
Runner identity & basic race-day fields (selectionId, runnerName, sometimes jockey/trainer/age/weight/barrier)	Betfair `listMarketCatalogue` with `RUNNER_METADATA` [1]	Only queryable prospectively for published markets; must snapshot pre-race because `listMarketCatalogue` targets `ACTIVE/SUSPENDED` markets [1]	"When available" keys; completeness must be measured in AU production [1]	**Medium**: personal vs commercial use is explicitly regulated via Betfair licensing; commercial usage requires approval/licensing [13]
Price/volume market history for backtests	Betfair historical stream data [14]	AU/NZ from Oct 2016 [7]	High fidelity for odds; runner metadata absent [15]	**Medium**: governed by Betfair licensing; treat as controlled data asset [13]
Horse run history (last N starts), scratchings/conditions updates, ratings, and sectionals (AU focus)	entity["company","Punting Form","racing data api	australia"] API (e.g., `/v2/form/form`) [16]	Form endpoint includes up to last 10 runs per horse [17]; marketing indicates 10+ years of data for modeller use and options for historical sectionals and modelling-ready point-in-time data [18]	API explicitly exists; sectionals available on higher tiers; point-in-time modelling data is explicitly referenced (critical for leakage control) [19]

Feature block	Source endpoint/provider	Historical depth	Missingness & latency	Licensing risk for model training
Cross-jurisdiction horse/jockey/trainer/course data, plus ratings content (global, including AU via B2B)	entity["company","Timeform","racing ratings provider	uk"] APIs/products (Timeform API; Global Sports API) [20]	Timeform API advertises access to races/cards/results and horse/trainer/jockey details, including historical data [21]	Global Sports API states it can deliver coverage of North America and Australia for “follow-the-sun” horse racing coverage [22]
Official fields/results material (authority record), race-day integrity artefact	entity["organization","Racing Australia","thoroughbred authority	aus"] web materials [23]	Long historical record in practice; official results/records are defined at the rules level [24]	Not an ML-ready API by default; ToU explicitly restricts use
TAB-form and fields feed via “TAB Web Services”	entity["company","Tabcorp","wagering company	australia"] “TAB Web Services” terms [26]	Not specified publicly in ToU	ToU frames it as a digital API; usage constrained
Third-party “developer-friendly” horse racing API with AU add-on	entity["company","The Racing API","horse racing data api	uk"] [27]	Claims large results DB and AU coverage options [27]	Updates described as periodic; suitability for point-in-time features must be validated

Legal and operational constraints that materially affect design

Betfair's developer documentation is explicit that application keys are for personal betting development, and commercial usage must be approved; unauthorised commercial usage can be blocked. [13] Betfair's broader terms also prohibit scraping/automated collection of Betfair data without consent, which directly impacts any "unsupported endpoint" strategy. [28]

For "official" racing materials, Racing Australia's website terms explicitly restrict use to personal, non-commercial purposes and prohibit automated scraping or reuse, making it unsuitable as the backbone of an automated training pipeline without a proper licence. [25]

If your roadmap includes any move toward an operational wagering product (as distinct from personal research), note that race fields information is regulated at state/territory level; a bookmaker must obtain "race field approval" from relevant governing bodies, and penalties exist for publishing/using race fields without approval. [29] This is not a modelling detail, but it changes how aggressively you can operationalise external "fields" data sources in a commercial setting. [29]

Entity resolution when IDs are inconsistent

A leakage-safe form layer fails most often at **identity joins**, not at model choice. The working assumption should be that **no single ID** spans Betfair selections, external form providers, and any authority data. Betfair's own runner metadata may include `runnerId` and rich breeding fields (sire/dam), but these are still "when available" and should not be treated as universally present in AU. [1]

A robust resolution strategy for horse, jockey, and trainer entities should be evidence-based and reversible:

Canonical entity store - Maintain `dim_horse`, `dim_jockey`, `dim_trainer` with stable internal IDs. - Maintain source-specific mapping tables: `map_betfair_selection`, `map_provider_horse`, `map_provider_jockey`, etc., storing (source_id, internal_id, confidence, evidence_fields, first_seen_ts, last_seen_ts).

Name normalisation and collision handling - Deterministic normalisation first: Unicode normalisation, punctuation stripping, whitespace collapse, casefold, and domain-specific suffix rules (e.g., "(NZ)", "(IRE)", "Jnr", "Snr") while preserving a reversible original form. - Fuzzy matching only as a second stage, with a conservative threshold and mandatory disambiguators: - Horse: (normalised name) + (foaling year/DoB if available) + (sex) + (sire/dam where present) + (trainer at time). - Jockey/trainer: (normalised surname) + (initials) + (jurisdiction/meeting context) and stable alias sets. - Explicit collision workflow: if two candidates tie within epsilon, assign **no automatic link**; create a "needs_review" record and fall back to higher-level aggregates (e.g., "unknown jockey") at feature time.

Point-in-time identity - Store "as-of" relationships (horse ↔ trainer, horse ↔ jockey for this start) as facts keyed by (race_id, runner_internal_id, asof_ts). Late rider changes are common enough that temporal versioning is required for no-lookahead.

Feature spec v1 with leakage proofs

A leakage-safe spec requires: (i) the precise formula, (ii) the required source tables, and (iii) an auditable proof that the feature is computable using only information available at **T-10 minutes**.

Notation: - Race start time: T_0. Feature cutoff: $T_c = T_0 - 10 \text{ minutes}$. - Horse h, jockey j, trainer t, race r, runner in today's field $i \in \{1, \ldots, n_r\}$. - Historical starts for horse h: $S_h = \{s : \text{start\_time}(s) < T_c\}$.

Horse recent form block

Last-N starts counts - N (default 5 or 10). - $\text{starts\_lastN}(h) = \min(N, |S_h|)$ - $\text{wins\_lastN}(h) = \sum_{s \in S_h^{(N)}} \mathbf{1}[\text{finish\_pos}(s) = 1]$ - $\text{places\_lastN}(h) = \sum_{s \in S_h^{(N)}} \mathbf{1}[\text{finish\_pos}(s) \le 3]$

Recency - $\text{days\_since\_run}(h) = \frac{T_c - \max_{s \in S_h} \text{start\_time}(s)}{1 \text{ day}}$ - If you use Betfair's `DAYS_SINCE_LAST_RUN`, treat it as a convenience field with missingness and still recompute from your own historical starts to ensure consistency. [1]

Exponentially decayed win/place rates - Choose half-life τ (e.g., 60 days). - Weight per start: $w_s = \exp\left(-\frac{T_c - \text{start\_time}(s)}{\tau}\right)$ - $\text{ewinrate}(h) = \frac{\sum_{s \in S_h} w_s \mathbf{1}[\text{finish\_pos}(s)=1]}{\sum_{s \in S_h} w_s}$

Distance / surface / track splits - Bucket distance into coarse bins (engineering-friendly): sprint / mile / middle / staying. - For a context key $k = (\text{surface}, \text{distance\_bin}, \text{track})$: - $\text{ewinrate}_k(h)$ computed as above but restricting s to those with the same k.

Class progression - Define a numeric class index $C(r)$ from the external source class label (e.g., by mapping Group/Listed/class/benchmark bands to a monotone scale). - $\Delta C(h) = C(r_{\text{today}}) - C(r_{\text{last start before } T_c})$

Leakage audit proof (horse form): all terms only reference starts with `start_time < T_c`. This is enforceable purely by time filtering during feature build; it does not require knowing result publication time. (Risk only arises if the source provides corrected results with retroactive edits; mitigate by snapshotting your upstream extracts or using provider "point-in-time" datasets when available.) [30]

Jockey and trainer blocks

Rolling jockey performance (time-windowed to avoid leakage via "future rides") - For a lookback window W days: $R_j = \{s : \text{jockey}(s) = j, \text{start\_time}(s) \in [T_c - W, T_c)\}$ - $\text{j\_rides}(j) = |R_j|$ - $\text{j\_winrate}(j) = \frac{\sum_{s \in R_j} \mathbf{1}[\text{finish\_pos}(s)=1]}{|R_j|}$ (with Bayesian smoothing recommended)

Rolling trainer performance analogously with R_t.

Interaction terms - Horse–jockey: compute on $R_{h,j} = \{s : \text{horse}(s) = h, \text{jockey}(s) = j, \text{start\_time}(s) < T_c\}$ - Jockey–trainer: $R_{j,t}$ similarly.

Leakage audit proof (jockey/trainer): only pre-T_c rides/runners contribute. Rider changes after T_c are handled by freezing the jockey field at the T-10 snapshot; this is necessary because rider data can be missing or change late in some feeds. [31]

Field-relative strength features

Field strength is often the highest-leverage "form intelligence" because it turns absolute ratings into a **within-race** signal and is naturally leakage-safe when based on pre-T_c ratings.

Given a per-runner rating $R_i(T_c)$:

- **Percentile:** $\mathrm{pct}_i = \frac{1}{n_r}\sum_j \mathbf{1}[R_j(T_c) \le R_i(T_c)]$
- **Gap to top:** $\Delta_i^{\mathrm{top}} = \max_j R_j(T_c) - R_i(T_c)$
- **Top-k strength:** $\mathrm{topk\_mean} = \frac{1}{k}\sum_{j\in\mathrm{TopK}} R_j(T_c)$
- **Competitiveness entropy:** with implied win probabilities p_i,
- $H = -\sum_i p_i \log p_i$ (higher $H \rightarrow$ more even race)

Leakage audit proof (field-relative): the field is defined as "active runners at T_c"; this must use a point-in-time runner set (including scratchings up to T_c). Scratchings after T_c are out of scope by design; optionally route such races to a "late-change" handling path.

Dynamic ratings (Elo-style + one alternative)

Horse racing is **multi-competitor**, so vanilla two-player Elo is not the cleanest formulation. The two most engineering-friendly rating families for racing are: (i) a multinomial "Elo-like" gradient update, and (ii) a Plackett–Luce model update using rankings. Plackett–Luce is a standard ranking model for permutations. (Plackett, 1975). [32]

Elo-style multinomial (winner-only; very stable) - Pre-race win probability:

$$p_i = \frac{\exp(R_i/s)}{\sum_{j=1}^{n_r}\exp(R_j/s)}$$

- Outcome indicator: $y_i = 1$ for winner else 0. - Update:

$$R_i \leftarrow R_i + K\,(y_i - p_i)$$

This retains the "proper direction" property of Elo (Elo, 1978) while avoiding pairwise explosion. [33]

Alternative: Plackett–Luce top-m update (uses more of the finish order) - For observed finish order π and top-m positions:

$$\log P(\pi) = \sum_{k=1}^{m}\left(R_{\pi_k} - \log\sum_{j\in S_k}\exp(R_j)\right)$$

where S_k is the set of runners not yet placed in positions $1..k-1$. - Gradient step update:

$$R_i \leftarrow R_i + \eta \frac{\partial}{\partial R_i} \log P(\pi) - \lambda R_i$$

This is grounded in the Plackett–Luce model literature and admits efficient inference/optimisation methods. [34]

Time decay (ratings drift) - Between starts for horse h, shrink toward baseline R_0:

$$R_h(T_c) = R_0 + (R_h(T_{\text{prev}}) - R_0) \exp(-\gamma \Delta \text{days})$$

This is critical in racing because layoff spells and changing stable conditions create non-stationarity.

Hierarchical extensions (recommended but staged) Represent a runner's effective rating as:

$$R^{\text{eff}} = R_{\text{horse}} + R_{\text{surface}} + R_{\text{distance\_bin}} + R_{\text{track}} + R_{\text{jockey}} + R_{\text{trainer}}$$

and update each component with shared shrinkage to avoid overfitting. This can be staged after the base horse rating is stable.

Experiment plan, metrics, and ablations

A leakage-safe evaluation must mirror live inference: features available at T_c, runner set frozen at T_c, and market price features sampled exactly at T_c (not SP/BSP unless you only bet at BSP).

Baselines and ablations

Baseline should reproduce your current production setup: market-only features and your existing routing/calibration. The ablation ladder should isolate incremental value:

1. **Baseline (market-only).**
2. **+ Betfair runner metadata block** (trainer/jockey/age/barrier/weight, etc.) where present. [6]
3. **+ Horse history form block** (last N starts, recency, splits).
4. **+ Jockey/trainer rolling stats + interactions.**
5. **+ Field-relative strength** computed from dynamic ratings.
6. **+ Dynamic ratings** (multinomial Elo first; PL top-m second).

Cross-validation and holdout protocol

Use **time-ordered evaluation** (blocked or rolling-origin) rather than random CV, because horse/jockey/trainer strengths drift and identity resolution errors can leak information across folds if the same horse appears in both train and test in close time proximity. This is a modelling constraint, not a preference.

Operationally: - Define a long training window, then evaluate on sequential monthly blocks (or weekly blocks) with a fixed embargo around the test window if you engineer any features that could be revised retroactively (e.g., corrected results, re-handicaps). - Group at the race level so all runners in a race belong to the same fold.

Metrics

For probability models, log loss (cross-entropy) and Brier score both directly measure probabilistic accuracy; ECE measures calibration quality:

- **Log loss / cross-entropy**: widely used as a proper scoring rule, optimised when predicted probabilities match true probabilities. [35]
- **Brier score**: mean squared error on probability forecasts (Brier, 1950; Murphy, 1986). [36]
- **ECE (Expected Calibration Error)**: defined via binning confidence and comparing empirical accuracy to predicted confidence (Guo et al., 2017). [37]
- **Cohen's kappa**: agreement statistic sometimes used for categorical decisions (Cohen, 1960). [38]

For betting robustness, evaluate expected value stability rather than only realised ROI: - At T_c, with available odds $o_i(T_c)$ and model win probability $\hat{p}_i$, the implied overlay is $\hat{p}_i - 1/o_i$. This mirrors the value calculation described in Betfair's own ratings model explanation. [39]
- Report performance by odds bands (e.g., favourites vs longshots) to prevent "ROI spikes" from a narrow segment masking calibration degradation.

Implementation blueprint with backfill and operational fallbacks

A leakage-safe "form intelligence" layer is primarily a **data versioning** problem: you must be able to rebuild features exactly as they were at T_c for any race in training and live.

Storage and schema (minimum viable)

1. **Raw snapshots (immutable, point-in-time)**
2. `raw_market_catalogue_snapshot(market_id, ts, json, projection_set)`
3. `raw_market_book_snapshot(market_id, ts, json, price_projection)`
4. Optional: `raw_external_racecard_snapshot(source, ts, race_key, json)`
5. **Conformed dimensions**
6. `dim_runner_internal(internal_runner_id, horse_internal_id, name_canonical, source_keys…)`
7. `dim_jockey_internal`, `dim_trainer_internal`
8. **Race entry fact (as-of T_c)**
9. `fact_race_entry(race_id, market_id, internal_runner_id, asof_ts=T_c, barrier, declared_weight, jockey_internal_id, trainer_internal_id, scratched_flag_at_Tc, …)`
10. **Feature store**

11. `feat_runner_win_v1(race_id, internal_runner_id, asof_ts, feature_vector, feature_version)`

Ingestion changes

- Schedule a **T-10 capture job** keyed off market start times:
- Pull `listMarketCatalogue` with `RUNNER_METADATA` and persist raw JSON. [1]
- Pull `listMarketBook` (your existing market pipeline) at T_c and persist.
- Optionally capture earlier snapshots (T-60, T-30) to diagnose late changes and improve robustness.

Because historic stream archives do not contain runner metadata, these snapshots are what make the feature layer reproducible and leakage-safe. [15]

Backfill strategy

- **Odds/market features:** backfill from Betfair historical stream files (AU since Oct 2016). [40]
- **Form/history features:** backfill from an external provider that supplies historical runs (e.g., Punting Form form endpoint includes last 10 runs) and/or explicitly provides modelling-ready point-in-time datasets (ideal). [41]
- **Runner metadata (jockey/trainer/weight/barrier at T_c):** not reliably backfillable from official Betfair historical archives; treat as prospective-only unless you have a separately licensed, timestamped racecard feed. [15]

Avoid building the core training set off “unsupported” Betfair Hub endpoints unless you have explicit permission and can guarantee stability; the same Betfair tutorial that documents them also frames them as unsupported, and Betfair’s terms restrict automated collection. [12]

Fallback behaviour when external feeds are missing

Implement explicit degradation paths rather than silent imputation: - If external form feed fails: compute model inputs from market-only + any available Betfair runner metadata (trainer/jockey/age/barrier/weight). [42]

- If jockey/trainer missing: use “unknown” category with pooled priors; do not attempt last-minute fuzzy matching without evidence. - If track condition is missing or only available as a post-race “final” value: drop track-condition-sensitive features for that race to avoid subtle leakage.

Runtime cost estimate (order-of-magnitude)

If you precompute rolling horse/jockey/trainer stats and dynamic ratings daily (or incrementally after each race result), live inference at T_c reduces to: - reading the T-10 snapshots, - assembling per-runner feature vectors, - computing field-relative aggregates (linear in number of runners), - scoring via the existing model service.

This is typically sub-second per race in a standard feature store + model server architecture, with the main latency driver being upstream API calls rather than feature computation.

Prioritised roadmap (uplift vs risk)

P0 (lowest risk, fastest measurable uplift) - Persist T-10 `listMarketCatalogue(RUNNER_METADATA)` snapshots and build missingness monitoring by key. [1]
- Implement multinomial Elo-style horse rating and field-relative strength features (percentiles, gaps, entropy) using only pre-T_c results. [43]
- Add conservative jockey/trainer rolling rates only when identity resolution confidence is high.

Expected effect: improved generalisation via field-strength normalisation; robust uplift in log loss/Brier if the baseline is market-only, with minimal leakage surface.

P1 (higher uplift, requires external licensing but still manageable) - Integrate a licensed form provider focused on AU (Punting Form) for last-10 runs and sectionals; explicitly prefer "point-in-time modelling data" if available to satisfy strict no-lookahead. [44]
- Add distance/surface/track splits and richer trainer/jockey interactions.

Expected effect: stronger lift for mid-price runners where market-only features underrepresent latent form edges; improved calibration in non-metro meetings if sectionals and run-style features are available.

P2 (complexity increase; target PLACE/EXOTIC portability) - Implement Plackett–Luce top-m updates (or a Bayesian PL variant) to better model full finishing order, enabling principled PLACE and exotic probability components. [45]
- Explore hierarchical rating decomposition (horse + context + jockey/trainer effects) with shrinkage.

Expected effect: best pathway to PLACE/exotics portability, but highest risk of brittle identity joins and overfitting without rigorous regularisation and long backtests.

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